

THE TWO-SPEED BRAIN ARCHITECTURE & PRIVILEGE BOUNDARY

Slow Deliberative System 2 (1–5 Hz) $\leftrightarrow$ Fast Real-Time System 1 (100–1000 Hz) with Proposal–Permission Gating

UNTRUSTED DELIBERATIVE COGNITIVE REALM · APPLICATION PROCESSOR (HOST LINUX MPU / ACCELERATOR, ~35W TDP)

AUTHORITY: UNPRIVILEGED CANDIDATE PROPOSALS ONLY

SYSTEM 2: COGNITIVE FOUNDATION MODEL / VLM

1 – 5 Hz · Latency: 200–1000 ms

- **Substrate:** Multi-modal Transformer (7B–70B weights) on NPU/GPU
- **Function:** Open-world semantic grounding, object affordance & goal logic
- **Artifact Emitted:** Expiring Intent Lease $L_{\text{intent}} = (G, V_{\text{scope}}, t_{\text{expire}})$
- **Failure Profile:** Semantic hallucinations, P99 tail spikes, OOD unreliability

Lease

SYSTEM 1.5: GENERATIVE TRAJECTORY PLANNER

20 – 50 Hz · Horizon $H=16$

- **Substrate:** Diffusion Policy / Action Chunking Transformer (ACT)
- **Function:** Synthesizes smooth C^2 continuous spline waypoints $A_{\{t:t+H\}}$
- **Artifact Emitted:** Candidate Action Chunk Proposal $p_t = \{q_k, \dot{q}_k, \tau_k\}$
- **Seam Behavior:** Recedes on arrival; decelerates to hold if next chunk delays

PROPOSAL–PERMISSION PRIVILEGE BOUNDARY (NO DIRECT ACTUATOR ACCESS)

Unverified Proposal p_t

TRUSTED REAL-TIME SAFETY REALM · DUAL LOCK-STEP MCU (BARE-METAL / FreeRTOS, < 1.5W TDP, 0 DYNAMIC MALLOC)

AUTHORITY: SOLE PERMISSION TO ENERGIZE ACTUATORS

SYSTEM 1: DETERMINISTIC BARRIER ENFORCER & SAFETY GATE

1000 Hz · Jitter < 5 μ s

- **Control Barrier Function:** Solves Active-Set QP: $\min \|u - p_t\|^2$ s.t. $\dot{h}(x) + \alpha(h(x)) \geq 0$
- **Dynamic Stopping Distance:** Evaluates $d_{\text{stop}}(v_t) = v_t \cdot t_{\text{react}} + v_t^2 / (2 a_{\text{max}}) \leq D_{\text{clear}}$
- **Temporal & Lease Check:** Rejects p_t if $t > t_{\text{expire}}$ or CRC/timing deadline missed
- **Dual Execution Verdict:** Passes safe command $u_t = \text{permit}(p_t)$ OR triggers Safe-Stop $\perp$

ACTUATION POWER STAGE & PHYSICAL BODY

Irreversible State ($W_t \rightarrow W_{\{t+1\}}$)

u_t (Safe)

- **Gate Driver Registers:** Direct H-Bridge gate drive, L/R electrical current rise
- **Electromagnetic Torque:** Delivers mechanical torque $\tau = k_t \cdot I$ into gearboxes
- **Physical Conservation Laws:** Momentum $p=mv$ · Joule Heat I^2R · Friction dissipation
- **Irreversible Boundary:** Once current energizes stator, mechanical motion is committed

$\perp$ (E-Stop)

THE BEDROCK PRINCIPLE OF THE TWO-SPEED ARCHITECTURE

High-level neural models propose rich, open-world intentions over extended horizons without physical privilege;
Hard real-time deterministic enforcers audit every proposal at microsecond cadences, retaining exclusive permission to energize motors.